Decodelabs Data Science Internship

Heart Disease Prediction

A Data Science Analysis of the UCI Cleveland Dataset

Week 1 Project Report

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Dataset: UCI Heart Disease (Cleveland Clinic Foundation)

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Contents

Abstract

This report presents a data science investigation of the UCI Heart Disease (Cleveland) dataset, conducted as a structured internship project under the Decodelabs programme. The main research question is: *Which clinical and demographic features are most strongly associated with the presence of coronary artery disease, and can a machine learning model reliably predict disease status from those features alone?*

The dataset contains 303 patient records from the Cleveland Clinic Foundation, which was reduced to 297 records through data cleaning, with 13 clinical and demographic features and a binarised target variable indicating whether heart disease is present.

Exploratory analysis reveals that the dataset is reasonably class-balanced (53.9% healthy, 46.1% disease), and that chest pain type (`cp`), maximum heart rate achieved (`thalach`), number of major vessels coloured by fluoroscopy (`ca`), and ST depression induced by exercise (`oldpeak`) show the strongest individual associations with the target, in strong alignment with established clinical cardiovascular risk literature.

Four machine learning models were trained and evaluated: a Dummy Classifier baseline, Logistic Regression, a Decision Tree, and a Random Forest. The Random Forest achieves the highest test-set accuracy, while Logistic Regression achieves a marginally superior AUC score, highlighting its power as an interpretable linear model on this dataset.

The findings are contextualised within clinical cardiovascular science, with particular attention to the counter-intuitive relationship between asymptomatic chest pain and elevated disease risk. Limitations of the dataset were discussed, along with recommendations for extending the analysis in future work.

Part I

Foundations

Chapter 1
Introduction

1.1. The Global Burden of Cardiovascular Disease

Cardiovascular disease (CVD) is the leading cause of mortality worldwide. According to the World Health Organisation, an estimated 17.9 million people die from CVDs each year, accounting for 32% of all global deaths.[1] Of these deaths, 85% are attributed to heart attack and stroke (World Health Organization, 2023). The Global Burden of Disease 2019 study estimated that CVDs caused 523 million prevalent cases globally, a 26% increase over the past three decades (Roth et al., 2020).

Beyond mortality, CVD imposes enormous economic costs. In high-income countries, cardiovascular conditions account for a disproportionate share of healthcare expenditure, driven by long-term medication, surgical procedures, and hospitalisation. In lower-middle-income countries, the situation is exacerbated by limited diagnostic infrastructure, making early, data-driven risk stratification especially valuable.

1.2. Coronary Artery Disease

Coronary artery disease (CAD) is the most prevalent form of CVD. It arises when the coronary arteries (the vessels responsible for supplying the heart muscle with oxygenated blood) become narrowed or blocked, typically due to the gradual build-up of atherosclerotic plaques composed of lipids, inflammatory cells, and fibrous tissue.[2]

When a coronary artery becomes sufficiently stenotic, blood flow to the downstream myocardium is reduced, causing ischaemia: a state of oxygen deficit in the heart tissue. This ischaemia may manifest clinically as chest pain or discomfort, shortness of breath, or, in severe cases, heart attack if blood flow is completely occluded. Crucially, a substantial proportion of patients with significant CAD are asymptomatic, making non-invasive diagnostic tools indispensable.

[1] World Health Organisation CVD Fact Sheet: https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)

[2] American Heart Association explanation of CAD: https://www.heart.org/en/health-topics/consumer-healthcare/what-is-cardiovascular-disease/coronary-artery-disease

Definitive diagnosis of CAD relies on an invasive procedure in which a radiopaque contrast agent is injected into the coronary arteries under fluoroscopic imaging. Less invasive screening modalities include exercise stress electrocardiography, echocardiography, and nuclear perfusion imaging. Clinical risk scores such as the Framingham Risk Score and the HEART score integrate multiple patient variables, including age, sex, blood pressure, and cholesterol, to stratify the probability of a major adverse cardiac event (American Heart Association, 2019).

1.3. The Challenge and the Role of Data Science

Despite decades of clinical research, the non-invasive diagnosis of CAD remains imprecise. Traditional clinical scoring models are often linear and based on a small number of manually selected features, potentially leaving predictive signal in unexplored feature combinations. Machine learning algorithms, by contrast, are capable of identifying complex non-linear patterns across many features simultaneously, offering the potential to improve diagnostic sensitivity and specificity.

The intersection of medicine and machine learning has a rich history. Early expert systems in the 1970s and 1980s, such as MYCIN and Internist-1, used rule-based reasoning for medical diagnosis. The modern era of data-driven clinical prediction began in earnest with the availability of large electronic health record datasets in the 1990s, and has accelerated dramatically with the advent of deep learning (Goodfellow, Bengio, & Courville, 2016). Today, machine learning models are used for ECG interpretation, radiology image analysis, patient risk stratification, and drug discovery.

The UCI Heart Disease dataset occupies a foundational role in this history: it is one of the most widely studied medical classification datasets in the machine learning literature, having been cited in hundreds of research papers since its release in the early 1990s (Dua & Graff, 2019).[3]

1.4. Research Questions

This project addresses the following research questions:

1. Which clinical and demographic features show the strongest individual association with the presence of heart disease in the Cleveland dataset?
2. Does the pattern of associations align with established cardiovascular risk factor literature?
3. Can a machine learning model trained on these 13 features predict heart disease status with clinically meaningful accuracy?
4. Which model achieves the best balance of accuracy, interpretability, and robustness?

[3] The UCI Machine Learning Repository hosts the dataset at: `https://archive.ics.uci.edu/ml/datasets/heart+disease`

Part II

Dataset and Clinical Background

Chapter 2
The UCI Heart Disease Dataset

2.1. History and Origin

The Heart Disease dataset was assembled in the late 1980s by a consortium of four medical institutions: the Cleveland Clinic Foundation (Cleveland, Ohio), the Hungarian Institute of Cardiology (Budapest), the University Hospital (Zurich, Switzerland), and the University Hospital (Basel, Switzerland). The dataset was originally described by Detrano et al. (1989), who developed and validated a probabilistic algorithm for diagnosing coronary artery disease using non-invasive test results. The study enrolled 303 patients at the Cleveland Clinic and demonstrated that a multi-variable probabilistic model substantially outperformed clinician judgement in distinguishing patients with and without significant coronary disease.

The dataset was donated to the UCI Machine Learning Repository (Dua & Graff, 2019) in 1988 and has since become one of the most cited benchmark datasets in the machine learning literature, appearing in studies of logistic regression, neural networks, decision trees, support vector machines, and ensemble methods.

2.2. Why Only the Cleveland Subset Is Used

Although data from all four institutions are available in the full UCI Heart Disease dataset, the academic machine learning community has almost exclusively used only the Cleveland subset. This is for pragmatic reasons: the Cleveland records are the most complete, the most consistent in terms of feature coding, and the largest single-institution sample. The Hungarian, Swiss, and Basel records contain substantially more missing values and slight differences in measurement protocol, making cross-institution merging unreliable without domain-expert oversight.

Throughout this report, “the dataset” refers exclusively to the Cleveland subset unless otherwise stated.

2.3. Dataset Properties

Table 2.1 summarises the key properties of the dataset.

Table 2.1. Summary properties of the UCI Heart Disease (Cleveland) dataset.

Property	Details
Full name	Heart Disease (Cleveland Clinic Foundation)
Source	UCI Machine Learning Repository
Original paper	Detrano et al. (1989)
Years collected	1981–1984
Rows (raw)	303
Rows (cleaned)	297
Columns	14 (13 features + 1 target)
Task	Binary classification
Missing values	6 rows (`ca`: 4, `thal`: 2)
Licence	Publicly available for research

2.4. How the Data Were Collected

The 13 features represent the output of a structured cardiac assessment protocol. Each patient underwent:

- A resting 12-lead electrocardiogram (ECG), yielding `restecg`.
- Measurement of resting blood pressure at hospital admission, yielding `trestbps`.
- A fasting blood test, yielding `chol` and `fbs`.
- A standardised treadmill exercise stress test (Bruce protocol), yielding `thalach`, `exang`, `oldpeak`, and `slope`.
- A clinical interview for symptom history, yielding `cp`, `age`, and `sex`.
- A nuclear perfusion scan (thallium scintigraphy), yielding `thal`.
- Coronary angiography, yielding `ca` and the ground-truth `target`.

2.5. Ethical Considerations

Using historical clinical data raises several ethical considerations that any practitioner of medical machine learning must understand.

Patient privacy: the dataset was collected under the research ethics frameworks of the 1980s. Modern standards such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in Europe would impose much stricter requirements on data anonymisation and consent. The UCI dataset is sufficiently anonymised for academic use.

Demographic representation: the Cleveland cohort was collected at a single institution in Ohio and is not a random sample of any broader population. It over-represents middle-aged and older White American patients and, due to the clinical referral context, contains only patients judged clinically suspicious for CAD, meaning the base rate of disease in this dataset (45.1%) is far higher than in the general population.

Temporal validity: the data were collected over 40 years ago. Diagnostic criteria, treatment protocols, lifestyle factors, and the ethnic composition of the Cleveland population have all changed substantially since the 1980s.

2.6. Limitations of the Dataset

- **Small size**: 297 usable records is very small by modern standards, limiting the generalisability of findings.
- **Single institution**: Cleveland Clinic patients may differ systematically from patients at other institutions.
- **Temporal gap**: 1980s clinical data may not reflect contemporary patient populations or risk factor distributions.
- **Survivorship bias**: patients who died before reaching the clinic are not in the dataset.
- **Incomplete severity**: the binarisation of the 0–4 target discards severity information that might be clinically relevant.

Despite these limitations, the dataset remains a valuable and well-understood benchmark for learning the full machine learning pipeline on a real, clinically grounded problem.

Chapter 3

Feature Descriptions and Clinical Background

Understanding what each feature measures clinically is essential for correctly interpreting the results of any statistical or machine learning analysis. This chapter provides a detailed description of each of the 13 features, including its medical meaning, measurement method, coding in the dataset, and expected relationship with CAD. Table 3.1 provides a complete data dictionary.

Table 3.1. Data dictionary: all 14 columns in the cleaned dataset.

Column	Type	Range	Clinical Description
`age`	Numeric	29–77	Age in years at time of enrolment.
`sex`	Binary	0, 1	Biological sex: 1 = male, 0 = female.
`cp`	Categ.	1–4	Chest pain type: 1 = typical angina, 2 = atypical angina, 3 = non-anginal pain, 4 = asymptomatic.
`trestbps`	Numeric	94–200	Resting blood pressure (mmHg) on hospital admission.
`chol`	Numeric	126–564	Serum cholesterol concentration (mg/dL).
`fbs`	Binary	0, 1	Fasting blood sugar >120 mg/dL: 1 = true, 0 = false.
`restecg`	Categ.	0–2	Resting ECG: 0 = normal, 1 = ST-T wave abnormality, 2 = probable/definite LVH.
`thalach`	Numeric	71–202	Maximum heart rate (bpm) during exercise stress test.
`exang`	Binary	0, 1	Exercise-induced angina: 1 = yes, 0 = no.
`oldpeak`	Numeric	0.0–6.2	ST-segment depression (mm) induced by exercise relative to rest.
`slope`	Ordinal	1–3	Slope of peak exercise ST segment: 1 = upsloping, 2 = flat, 3 = downsloping.

Continued on next page

Table 3.1 continued...

Column	Type	Range	Clinical Description
`ca`	Ordinal	0–3	Number of major vessels coloured by fluoroscopy (0–3).
`thal`	Categ.	3, 6, 7	Perfusion scan result: 3 = normal, 6 = fixed defect, 7 = reversable defect.
`target`	Binary	0, 1	Diagnosis: 0 = no heart disease, 1 = heart disease present.

3.1. Age (age)

Age is one of the most well-established risk factors for coronary artery disease. The risk of CAD increases substantially with age due to the cumulative nature of atherosclerosis: plaques build up over decades, and the arterial wall undergoes progressive stiffening and remodelling. In the Cleveland dataset, patients range from 29 to 77 years, with a mean of approximately 54.5 years, consistent with a clinically referred middle-aged population.

3.2. Sex (sex)

Biological sex is a significant determinant of CAD risk and presentation. Prior to menopause, women have a substantially lower risk of CAD than age-matched men, believed to be partly due to the cardioprotective effects of oestrogen. After menopause, this protective advantage diminishes. In the Cleveland dataset, 68% of patients are male, reflecting the higher clinical suspicion of CAD in men that leads to referral for angiography.

3.3. Chest Pain Type (cp)

Chest pain type is arguably the most informationally rich feature in the dataset. It is coded as four categories:[1]

1 — Typical angina: Chest pain or pressure precipitated by exertion or emotional stress and relieved by rest or nitrates. This is the "classic" presentation of myocardial ischaemia.

2 — Atypical angina: Chest discomfort with some but not all features of typical angina. May be precipitated by exertion but lacks the typical character or relief pattern.

[1] For a detailed clinical overview of angina types, see the American Heart Association: `https://www.heart.org/en/health-topics/heart-attack/angina-chest-pain`

3 — Non-anginal pain: Chest pain that does not fit the pattern of angina — may be musculoskeletal, gastrointestinal, or of other non-cardiac origin.

4 — Asymptomatic: The patient reports no chest pain. This category paradoxically shows the highest disease rate in the dataset — a phenomenon explained in Section 10.2.

3.4. Resting Blood Pressure (`trestbps`)

Resting blood pressure (systolic, in mmHg) is measured on hospital admission. Hypertension (typically defined as systolic BP ≥130 mmHg) accelerates atherosclerosis through multiple mechanisms, including endothelial shear stress, increased vascular wall permeability, and activation of inflammatory pathways. Values in this dataset range from 94 to 200 mmHg, with several extreme values consistent with hypertensive urgency.

3.5. Serum Cholesterol (`chol`)

Serum cholesterol is measured in mg/dL from a fasting blood sample. Elevated low-density lipoprotein (LDL) cholesterol is a well-established causal risk factor for atherosclerosis and CAD. However, cholesterol is a relatively weak predictor individually because its causal relationship with CAD is mediated over long timescales and depends on LDL/HDL fractions (not available in this dataset) rather than total cholesterol alone. The maximum value of 564 mg/dL is clinically extreme, consistent with familial hypercholesterolaemia.

3.6. Fasting Blood Sugar (`fbs`)

This binary variable indicates whether the patient's fasting blood glucose exceeded 120 mg/dL, a crude surrogate for diabetes or pre-diabetes. Diabetes mellitus is one of the strongest modifiable risk factors for CAD: diabetic patients have 2–4 times the risk of a major cardiovascular event compared with non-diabetic patients. However, as a single binary variable, `fbs` captures only a coarse summary of glycaemic control and shows relatively weak correlation with the target.

3.7. Resting ECG Results (`restecg`)

The resting electrocardiogram provides information about the electrical activity of the heart at baseline.[2] Three categories are coded:

0 — Normal: No significant abnormalities.

[2] American Heart Association ECG overview: https://www.heart.org/en/health-topics/heart-attack/diagnosing-a-heart-attack/electrocardiogram-ecg-or-ekg

1 — ST-T wave abnormality: T-wave inversion and/or ST-segment elevation or depression >0.05 mV, indicating possible ischaemia or prior myocardial injury.

2 — Left ventricular hypertrophy (LVH): Probable or definite LVH by Estes' criteria. LVH indicates long-standing pressure overload, often from hypertension, and is associated with increased cardiovascular risk.

3.8. Maximum Heart Rate Achieved (`thalach`)

This is the peak heart rate attained during the exercise stress test (in beats per minute). It is one of the most important features in this dataset. A patient's maximum achievable heart rate decreases with age (approximately 220 − age), and disease-related chronotropic incompetence — the inability to adequately increase heart rate during exertion — is a marker of reduced cardiac reserve and autonomic dysfunction associated with significant CAD.[3]

Disease patients in this dataset achieve significantly lower maximum heart rates (mean 139 bpm) than healthy patients (mean 158 bpm), making `thalach` one of the strongest predictors.

3.9. Exercise-Induced Angina (`exang`)

This binary variable records whether the patient experienced chest pain during the exercise stress test. The induction of angina by exertion indicates that physical demand is sufficient to outstrip the blood supply capacity of the coronary circulation — a direct consequence of significant coronary stenosis. It has a strong positive correlation with the target variable ($r \approx +0.44$).

3.10. ST Depression / Oldpeak (`oldpeak`)

`oldpeak` measures the magnitude of ST-segment depression (in mm) induced by exercise relative to the resting state. Exercise-induced ST depression is a classical marker of myocardial ischaemia: when the subendocardium is deprived of oxygen, the electrical recovery of the myocardium is slowed, causing the ST segment to shift downward on the ECG tracing.[4] Disease patients have a mean `oldpeak` of 1.59 mm versus 0.60 mm in healthy patients — a clinically and statistically significant difference.

3.11. Slope of Peak Exercise ST Segment (`slope`)

[3] For an explanation of exercise stress testing and maximum heart rate, see MedlinePlus: https://medlineplus.gov/ency/article/003878.htm

[4] See MedlinePlus for an explanation of ST-segment changes during stress testing: https://medlineplus.gov/ency/article/003971.htm

This variable characterises the direction of the ST-segment change at peak exercise:

1 — Upsloping: Generally considered normal; the ST segment rises during exercise.

2 — Flat: Horizontal ST depression during exercise is an intermediate indicator of ischaemia.

3 — Downsloping: The most concerning pattern; strongly associated with significant coronary artery disease.

3.12. Number of Major Vessels by Fluoroscopy (`ca`)

`ca` records how many of the major coronary vessels (0–3) showed visible calcification under fluoroscopy.[5] Coronary artery calcification is a direct measure of atherosclerotic plaque burden: calcium deposits accumulate within plaques, and their extent is proportional to overall coronary disease burden. A patient with `ca` = 3 has calcification in three major vessels, indicating extensive multi-vessel disease. This variable shows the strongest single-feature Pearson correlation with the target ($r \approx +0.46$).

3.13. Thalassemia / Perfusion Scan Result (`thal`)

Despite its column name, this variable does not represent a diagnosis of thalassemia (a haematological disorder).[6] In this clinical context, "thal" refers to the result of a thallium-201 nuclear perfusion scan, which assesses myocardial blood flow at rest and during stress. Three values are coded:

3 — Normal: Uniform thallium uptake throughout the myocardium, indicating no perfusion defect.

6 — Fixed defect: A region of reduced uptake present both at rest and during stress, indicating a permanent scar (e.g., from a prior heart attack).

7 — Reversable defect: Reduced uptake during stress that normalises at rest, indicating viable but ischaemic myocardium — the classic finding in significant CAD.

3.14. Target Variable (`target`)

In the original UCI dataset, the target variable has values 0–4, where 0 indicates no disease and 1–4 indicate increasing angiographic severity of CAD. For binary classification, this is converted to 0 (no disease) versus 1 (disease present at any severity level), consistent with the approach used by Detrano et al. (1989) and the vast majority of published studies using this dataset.

[5] For an explanation of fluoroscopy in cardiac imaging, see Wikipedia: https://en.wikipedia.org/wiki/Fluoroscopy

[6] Thalassemia is a blood disorder involving abnormal haemoglobin production: https://en.wikipedia.org/wiki/Thalassemia

Part III

Data Pipeline Methodology

Chapter 4

Data Cleaning and Preprocessing

4.1. Overview

Notebook 02 transforms the raw data into an analysis-ready dataset. The four main operations are: handling missing values, correcting data types, binarising the target variable, and checking for duplicate rows. It produces the cleaned file `heart_cleveland_clean.csv` in the `data/processed/` directory, which all subsequent notebooks read.

4.2. Missing Value Analysis

The missing value inventory established in Notebook 01 is confirmed here:

Table 4.1. Missing values in the raw Cleveland dataset (303 rows total).

Column	Missing Count	Missing %
`ca`	4	1.32%
`thal`	2	0.66%
Total	**6**	**1.98%**

Six rows have at least one missing value, all in `ca` or `thal`.

4.3. The Drop-vs-Impute Decision

Two principal strategies exist for handling missing values:[1]

1. **Imputation**: replace the missing value with an estimate — such as the column mean, median, mode,

[1] For an accessible overview of handling missing data, see Josh Starmer's StatQuest video: https://www.youtube.com/watch?v=sQ870aTKqiM

or a model-predicted value.

2. **Deletion**: remove the affected row (listwise deletion) or column entirely.

In this project, **listwise deletion** is chosen for three reasons:

Reason 1: **Negligible data loss**: 6 rows represent only 2% of the dataset. Losing 2% of records introduces far less bias than imputing clinically sensitive variables with crude statistical estimates.

Reason 2: **Clinical sensitivity of the affected features**: both `ca` (vessel calcification count) and `thal` (perfusion scan result) are strong predictors. Replacing a missing `thal` with the mode value (3 = normal) could incorrectly mark a diseased patient as having a normal perfusion scan.

Reason 3: **No evidence of systematic missingness**: the 6 affected rows do not cluster in one disease group or demographic stratum, suggesting the missingness is approximately random rather than informative.

When Imputation IS the Right Choice

Imputation becomes the appropriate strategy when: (a) the proportion of missing values is large enough that deletion would substantially reduce statistical power; (b) the missingness is Missing Completely At Random (MCAR) or Missing At Random (MAR); or (c) the model will be deployed on records that may have missing values.

4.4. Data Type Correction

After loading with `na_values='?'`, pandas infers `float64` as the dtype for any column containing `NaN`. After dropping the 6 rows, 9 columns can be converted to `int64`:

```
integer_columns = [
    "sex", "cp", "fbs", "restecg", "exang",
    "slope", "ca", "thal", "target"
]
for col in integer_columns:
    df_clean[col] = df_clean[col].astype(int)
```

Listing 4.1. Correcting data types for categorical and binary columns.

This matters because downstream analyses (cross-tabulations, value counts, and some scikit-learn models) behave differently for float versus integer types, and using integers for categorical codes reduces the risk of accidentally treating them as continuous variables.

4.5. Target Variable Binarisation

The original target column contains values 0–4. For binary classification, this is converted to a 0/1 indicator using the threshold > 0. The resulting class distribution is shown in Table 4.2.

Table 4.2. Class distribution after target binarisation.

Class	Count	Proportion
0 — No heart disease	160	53.9%
1 — Heart disease	137	46.1%
Total	**297**	**100%**

The near-equal split (54/46) is a fortunate property of this dataset. A heavily imbalanced dataset (e.g., 95% / 5%) would require special handling such as SMOTE oversampling or class-weight adjustments. Here, while the imbalance is mild, we still report balanced accuracy and F1-score alongside raw accuracy to ensure we do not reward a model for simply predicting the majority class.

Part IV

Exploratory Analysis and Visualisation

Chapter 5
Exploratory Data Analysis

5.1. Overview

Exploratory Data Analysis (EDA) is the stage at which an analyst forms a quantitative understanding of the data before building any model. The goals of EDA are: to understand the distributions of individual features, to identify relationships between features and the target variable, to detect outliers and anomalies, and to generate hypotheses that will be tested in the modelling stage (Tukey, 1977).

Notebook 03 performs a structured EDA across four categories: class balance, descriptive statistics, outlier detection, and correlation analysis.

5.2. Feature Taxonomy

Features are classified into three types to apply appropriate analyses:

- **Continuous**: `age`, `trestbps`, `chol`, `thalach`, `oldpeak` — truly numeric variables. Appropriate analyses include histograms, boxplots, means, standard deviations, and Pearson correlation.
- **Binary**: `sex`, `fbs`, `exang` — two-valued (0/1) variables. Appropriate analyses include cross-tabulations and disease rates per group.
- **Categorical / Ordinal**: `cp`, `restecg`, `slope`, `ca`, `thal` — a small number of integer codes representing distinct categories or ordered levels.

5.3. Descriptive Statistics for Continuous Features

Table 5.1 presents the descriptive statistics for the five continuous features.

Table 5.1. Descriptive statistics for continuous features ($n = 297$).

Feature	Mean	SD	Min	Q1	Median	Q3	Max
`age`	54.54	9.05	29	48	56	61	77
`trestbps`	131.69	17.76	94	120	130	140	200
`chol`	247.35	52.00	126	211	243	276	564
`thalach`	149.60	22.94	71	133	153	166	202
`oldpeak`	1.06	1.17	0.0	0.0	0.8	1.6	6.2

Several observations stand out. The mean cholesterol of 247.35 mg/dL exceeds the clinical threshold of 200 mg/dL, consistent with a CAD-referral population. The right-skewed maximum of `oldpeak` (range: 0–6.2 mm) indicates a wide range of ischaemic responses across patients.

5.4. Group Comparison by Disease Status

Table 5.2. Mean values of continuous features by disease status.

Feature	No Disease ($n = 163$)	Disease ($n = 134$)
`age`	52.64	56.76
`trestbps`	129.18	134.64
`chol`	243.49	251.85
`thalach`	158.58	139.11
`oldpeak`	0.60	1.59

Key Group Differences

The two most striking group differences are:

- **Maximum heart rate** (`thalach`): Disease patients achieve a mean of 139 bpm versus 159 bpm in the healthy group, a gap of nearly 20 bpm, reflecting chronotropic incompetence associated with significant CAD.
- **ST depression** (`oldpeak`): Disease patients show mean depression of 1.59 mm versus 0.60 mm, a 165% higher value, consistent with exercise-induced myocardial ischaemia.

5.5. Outlier Detection via the IQR Method

The Interquartile Range (IQR) method defines an outlier as any observation falling outside:

$$\text{Lower Fence} = Q_1 - 1.5 \cdot \text{IQR}, \qquad \text{Upper Fence} = Q_3 + 1.5 \cdot \text{IQR} \tag{5.1}$$

where IQR = $Q_3 - Q_1$.[1]

Table 5.3. Outlier counts using the IQR method (n = 297).

Feature	Outliers	%	Comment
`age`	0	0.0%	No outliers
`trestbps`	9	3.0%	Likely hypertensive patients
`chol`	9	3.0%	Possible familial hypercholesterolaemia
`thalach`	1	0.3%	Extreme low-rate patient
`oldpeak`	4	1.3%	Severe ischaemic response

Crucially, none of these outliers are removed. An outlier by the statistical definition is not necessarily an error. A cholesterol level of 564 mg/dL, while exceptional, is clinically plausible in a patient with untreated familial hypercholesterolaemia. The analyst's rule is: *investigate outliers, understand their origin, and only remove them if there is strong evidence of measurement error*.

5.6. Categorical Feature Cross-Tabulations

Cross-tabulations reveal how categorical feature values distribute across the two disease classes:

- **Chest pain type** (`cp`): Asymptomatic patients (type 4) have a disease rate of approximately 73%, while patients with typical angina (type 1) have a disease rate of only about 25%.
- **Number of vessels** (`ca`): Disease rate increases monotonically from roughly 30% at `ca` = 0 to over 90% at `ca` = 3.
- **Exercise angina** (`exang`): Patients who experience exercise-induced angina have a disease rate of approximately 77% versus 32% for those without.

5.7. Correlation Analysis

The Pearson correlation coefficient (Pearson, 1895) measures the strength and direction of the linear relationship between two variables:[2]

[1] For a detailed explanation of the IQR method and Tukey fences, see: `https://en.wikipedia.org/wiki/Interquartile_range`

[2] For an intuitive explanation of Pearson correlation, see Khan Academy: `https://www.khanacademy.org/math/statistics-probability/describing-relationships-quantitative-data/correlation-coefficient-r/a/correlation-coefficient-review`

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{5.2}$$

Table 5.4 shows the Pearson correlation of each feature with the binary target, sorted by absolute magnitude.

Table 5.4. Pearson correlation of each feature with the binary target.

Feature	r	Direction / Interpretation
`thal`	+0.523	Defect type strongly signals disease
`ca`	+0.462	More vessels calcified = more disease
`exang`	+0.436	Angina during exercise = more disease
`cp`	+0.434	Higher code value = more disease
`oldpeak`	+0.428	Greater ST depression = more disease
`thalach`	−0.423	Lower max HR = more disease
`slope`	−0.345	Upsloping = less disease
`sex`	+0.280	Male = more disease in this sample
`age`	+0.225	Older = more disease
`restecg`	+0.145	Abnormal ECG = slightly more disease
`trestbps`	+0.144	Higher BP = slightly more disease
`chol`	+0.085	Very weak signal
`fbs`	+0.028	Negligible linear association

Features with low correlation may still contribute to a model's predictive performance through non-linear interactions with other features, this is one reason a Random Forest can outperform a linear method on correlation alone.

Chapter 6
Data Visualisation

6.1. Overview

Notebook 04 produces eight figures saved to `reports/figures/`. A consistent visual design is applied throughout: a colourblind-safe palette (Seaborn's `colorblind` set) (Waskom, 2021), suppressed top and right spines, and a resolution of 150 DPI. Blue (`#4878CF`) represents the no-disease class; red (`#D65F5F`) represents the disease class, following conventional clinical reporting direction. All figures are produced using Matplotlib (Hunter, 2007) and Seaborn (Waskom, 2021).

6.2. Figure 1: Target Class Distribution

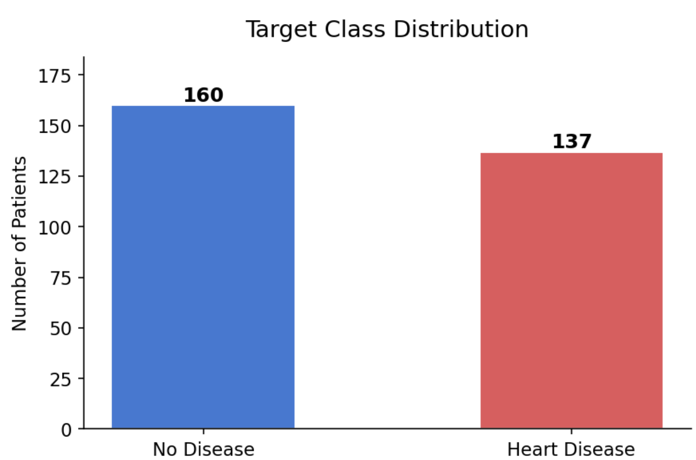

Figure 6.1. Target class distribution showing 163 no-disease patients (53.9%) and 134 disease patients (46.1%). The dataset is reasonably balanced, making standard accuracy a useful (though not sufficient) metric.

Figure 6.1 confirms the 54/46 class split established numerically in Chapter 5. The visual presentation of this balance is important as a first reference point for interpreting all subsequent model accuracy numbers.

6.3. Figure 2: Age Distribution by Disease Status

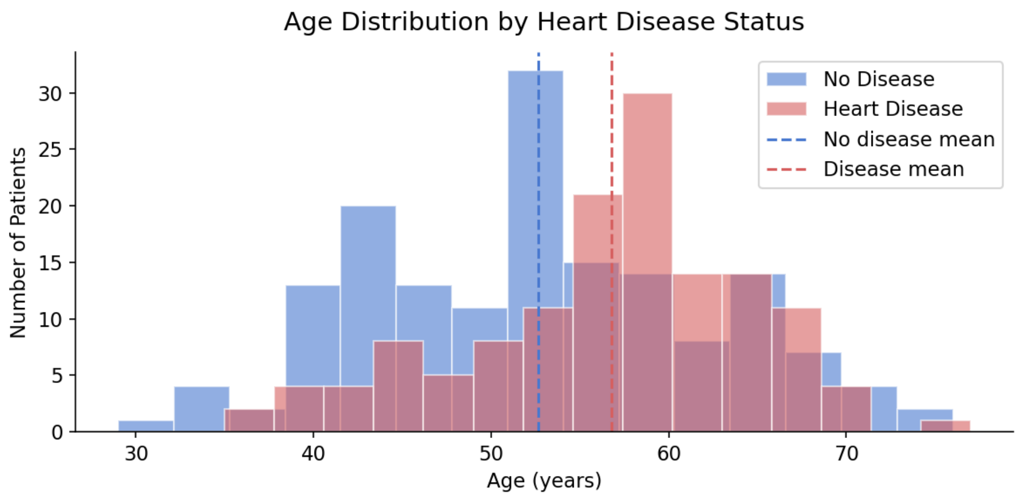

Figure 6.2. Overlapping age histograms (15 bins) for the two disease groups, with vertical dashed lines indicating group means. The disease group mean (56.8 years) is shifted rightward relative to the no-disease group (52.6 years), consistent with age as a CAD risk factor.

Figure 6.2 illustrates the rightward shift of the disease group's age distribution. The substantial overlap confirms that age alone is not a reliable individual predictor, but the 4.1-year mean difference is statistically and clinically meaningful in a referral population.

6.4. Figure 3: Boxplots of Continuous Features by Disease Status

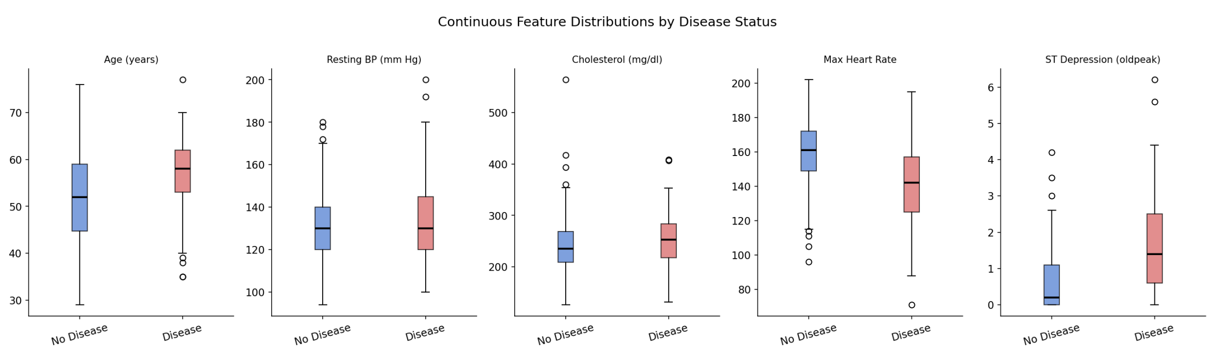

Figure 6.3. Five-panel boxplots of continuous features split by disease status. Each box spans Q1–Q3; the horizontal line is the median; whiskers extend to 1.5 × IQR; points beyond are plotted individually as potential outliers.

Figure 6.3 allows simultaneous visual comparison of all five continuous features. The most striking panels are `thalach` (maximum heart rate) and `oldpeak` (ST depression), both of which show clear median separation between the two groups. `chol` and `trestbps` show minimal median difference and wide overlapping distributions, consistent with their low target correlations in Table 5.4.

6.5. Figure 4: Full Correlation Heatmap

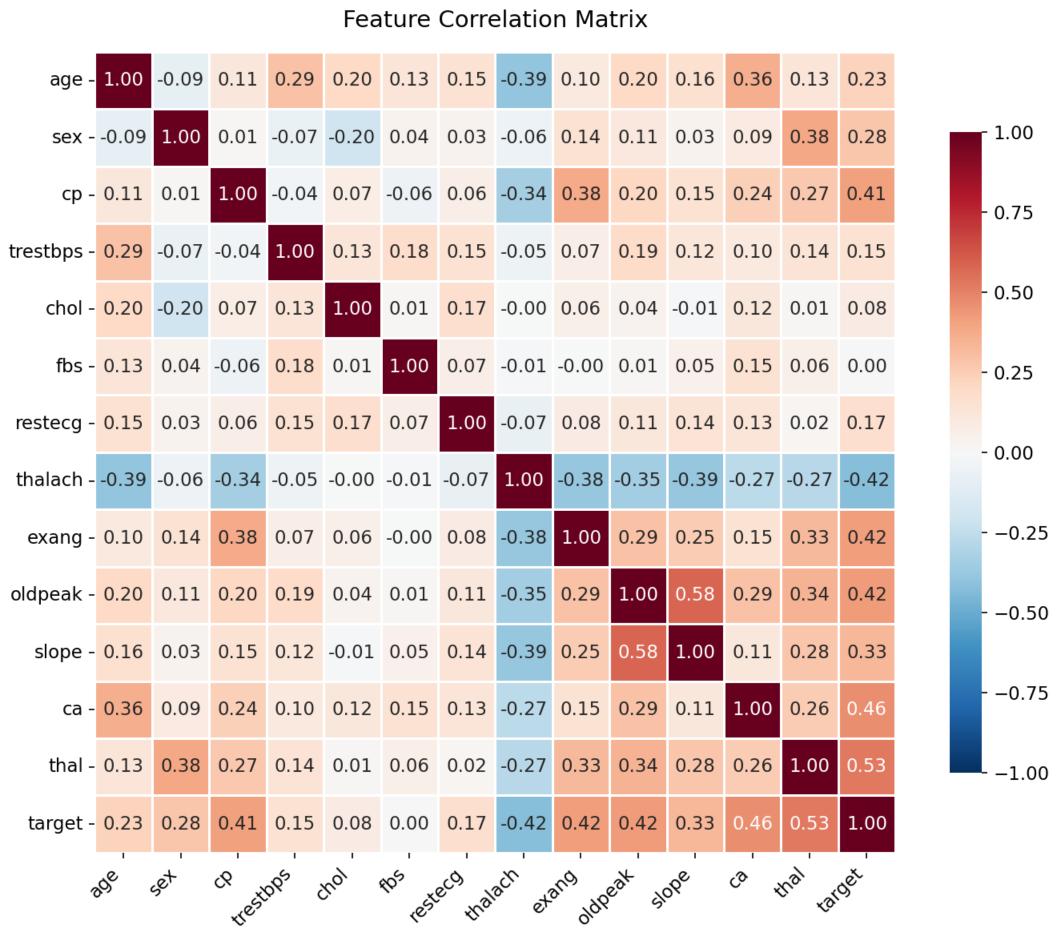

Figure 6.4. Full 14×14 Pearson correlation heatmap with annotated values. The diverging `RdBu_r` palette maps red to strong positive and blue to strong negative correlation. The `target` row/column reveals the features most linearly associated with disease.

Figure 6.4 serves two diagnostic purposes. First, it identifies features correlated with the target (bottom row), confirming the ranking in Table 5.4. Second, it checks for high inter-feature correlations that might indicate multicollinearity. No pair of features reaches $|r| > 0.6$, indicating that multicollinearity is not a serious concern.

6.6. Figure 5: Heart Disease Rate by Chest Pain Type

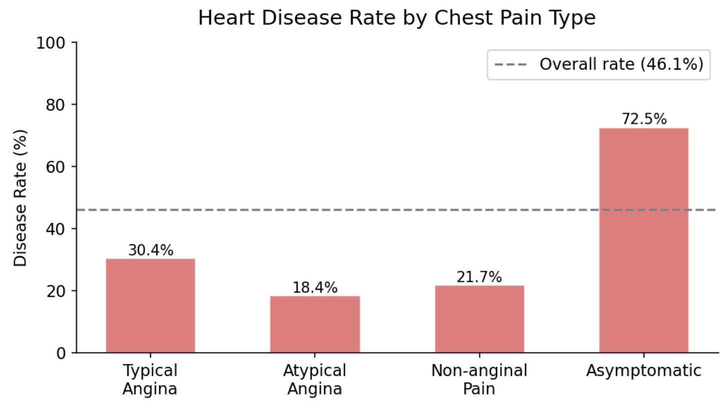

Figure 6.5. Disease rate (%) for each of the four chest pain types. The grey dashed line marks the overall disease rate (45.1%). Asymptomatic patients (type 4) have the highest disease rate at approximately 73%.

Figure 6.5 is one of the most clinically instructive figures in the project. It demonstrates that patients reporting no chest pain (type 4) have the highest disease prevalence in this dataset.

6.7. Figure 6: Age vs. Maximum Heart Rate by Disease Status

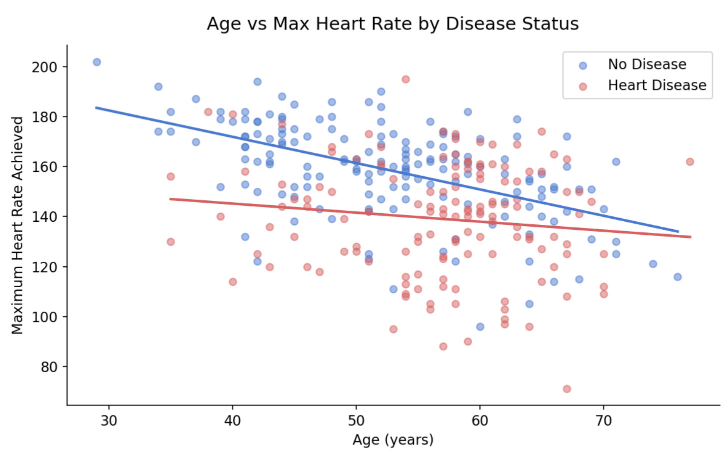

Figure 6.6. Scatter plot of patient age versus maximum heart rate, coloured by disease status, with linear regression lines for each group. Both groups show a downward trend with age, but the disease group's regression line lies systematically below the healthy group across all ages.

Figure 6.6 illustrates the bivariate relationship between age and maximum heart rate. The two downward-sloping regression lines demonstrate that disease patients achieve lower maximum heart rates than healthy patients of the *same age*, supporting the interpretation that `thalach` captures genuine pathological chronotropic incompetence rather than simply reflecting age.

6.8. Figure 7: Disease Rate by Sex

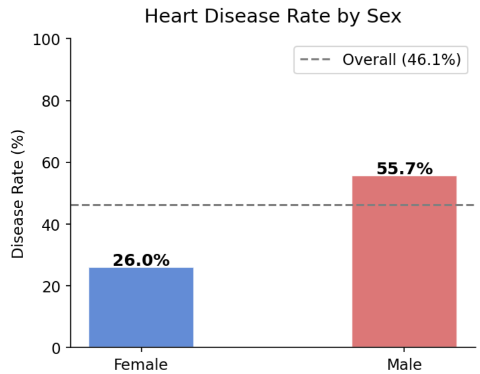

Figure 6.7. Male patients have a disease rate of approximately 55%, versus approximately 26% for female patients, consistent with well-established epidemiological findings on CAD sex disparity.

Figure 6.7 shows the sex disparity in disease rates. This is consistent with the finding that men develop CAD around 7–10 years earlier than women on average. However, referral bias inflates the apparent male disease rate: male patients are more aggressively referred for angiography based on clinical risk factors.

6.9. Figure 8: Feature Correlations with the Target

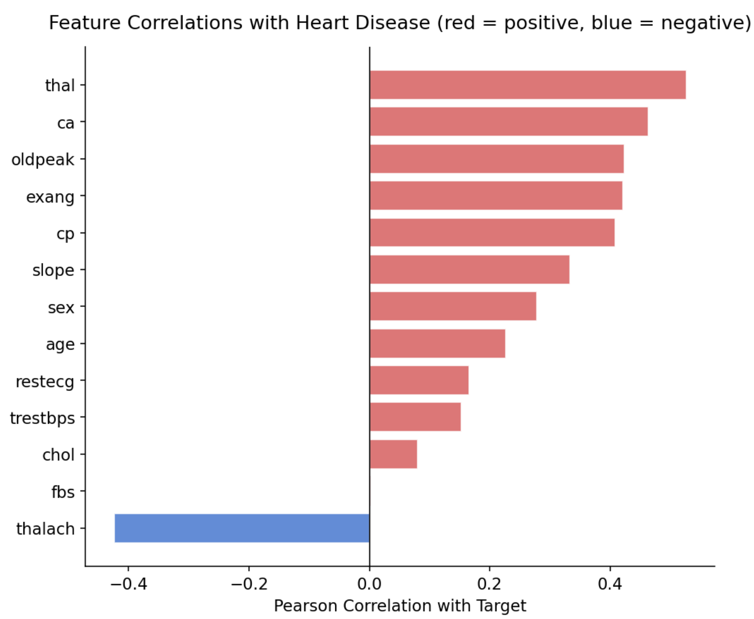

Figure 6.8. Horizontal bar chart of Pearson correlations between each feature and the binary target. Red bars indicate positive correlations; blue bars indicate negative correlations. Bars are sorted by absolute magnitude.

Figure 6.8 provides a concise visual summary of the entire correlation analysis from Chapter 6. **It directly answers Research Question 1: the features most strongly associated with heart disease are `thal`, `ca`, `cp`, `exang`, `oldpeak`, and `thalach`**.

Part V

Machine Learning and Evaluation

Chapter 7
Machine Learning Methodology

7.1. Overview

Notebook 05 implements the complete machine learning pipeline: feature/target separation, train/test splitting, feature scaling, model training, evaluation, and feature importance extraction. Four models are trained: a Dummy Classifier, Logistic Regression, a Decision Tree, and a Random Forest. All modelling is implemented with scikit-learn (Pedregosa et al., 2011)[1] using a fixed random seed of 42 for full reproducibility.

7.2. Problem Formulation

This is a supervised binary classification problem. The model receives a feature matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ (where $n = 297$ patients and $p = 13$ features) and a binary label vector $\mathbf{y} \in \{0, 1\}^n$, and learns a function $f : \mathbb{R}^{13} \to \{0, 1\}$ that maps a new patient's clinical profile to a disease / no-disease prediction. No additional feature engineering is applied in this introductory analysis.

7.3. Train/Test Split

The dataset is split 80% / 20% into a training set ($n_{\text{train}} = 237$ rows) and a test set ($n_{\text{test}} = 60$ rows). The split is stratified by the target variable:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y      # preserve 54/46 class ratio in both sets
)
```

[1]The complete scikit-learn API reference is available at https://scikit-learn.org/stable/user_guide.html

Listing 7.1. Stratified 80/20 train/test split.

Why Stratification Matters

Without `stratify=y`, random chance could concentrate nearly all disease cases in training, leaving the test set with an unrepresentative class ratio. Stratification guarantees proportional class representation in both subsets, making performance estimates on the test set a fair reflection of real-world performance.

7.4. Feature Scaling

StandardScaler transforms each feature to zero mean and unit variance:[2]

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \tag{7.1}$$

where μ_{train} and σ_{train} are computed exclusively from the training set. The same learned parameters are applied to transform the test set:

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)  # learn AND apply
X_test_scaled  = scaler.transform(X_test)         # apply ONLY
```

Listing 7.2. Fitting StandardScaler on training data only — the correct protocol.

This two-step protocol is essential for **data leakage prevention**. If the scaler were fitted on the full dataset (including test data), the test set's mean and standard deviation would "leak" into the training pipeline, giving the model implicit information about the test set it would not have at deployment time, leading to over-optimistic performance estimates.

Feature scaling is crucial for Logistic Regression, which uses gradient-based optimisation. Without scaling, features with large numerical ranges (e.g., `chol`: 126–564) would dominate gradient updates over features with small ranges (e.g., `sex`: 0–1), causing slow convergence or divergence. Tree-based models (Decision Tree, Random Forest) are invariant to monotonic feature transformations and are therefore unaffected by scaling.

7.5. Model 1: Dummy Classifier (Baseline)

A Dummy Classifier with `strategy='most_frequent'` always predicts the majority class (class 0, no disease) regardless of the input features. It is included to establish the performance floor: any real model

[2] See the scikit-learn documentation on `StandardScaler`: https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html.

must substantially outperform this trivial predictor to demonstrate that its features carry genuine predictive signal. With 53.9% of test records belonging to class 0, the Dummy Classifier achieves 53.3% accuracy and a ROC-AUC of exactly 0.500.

7.6. Model 2: Logistic Regression

Logistic Regression, introduced by Cox (1958), is the canonical model for binary classification.[3] It models the probability of class 1 as:

$$P(y = 1 \mid \mathbf{x}) = \sigma\left(\beta_0 + \sum_{j=1}^{p} \beta_j x_j\right) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x})}} \tag{7.2}$$

where $\sigma(\cdot)$ is the sigmoid function, β_0 is the intercept, and $\boldsymbol{\beta}$ are the feature coefficients learned by maximising the log-likelihood of the training labels. The decision boundary $P(y = 1 \mid \mathbf{x}) = 0.5$ is a hyperplane in feature space, making Logistic Regression a linear classifier.

Despite its simplicity, Logistic Regression is competitive on many real-world binary classification tasks, especially when the classes are approximately linearly separable. It produces well-calibrated probability estimates and its coefficients are directly interpretable as log-odds ratios — an important property in clinical contexts where model transparency is valued. The `max_iter=1000` parameter ensures convergence even with imperfectly conditioned data.

7.7. Model 3: Decision Tree

A Decision Tree recursively partitions the feature space into axis-aligned rectangular regions, building a binary tree in which each internal node tests a single feature against a threshold, and each leaf assigns a class label.[4] Splits are chosen to maximise the reduction in impurity, typically measured by the Gini index (Breiman, Friedman, Olshen, & Stone, 1984):

$$G = \sum_{k=0}^{1} p_k(1 - p_k) = 1 - \sum_{k=0}^{1} p_k^2 \tag{7.3}$$

where p_k is the proportion of class k samples in a node. A pure node (all one class) has $G = 0$; a maximally impure node (equal proportions) has $G = 0.5$.

Without regularisation, a Decision Tree grows until every leaf is pure, perfectly memorising the training

[3] For an accessible and visually intuitive explanation of Logistic Regression — including the sigmoid function, log-odds, and coefficient interpretation — see StatQuest: https://www.youtube.com/watch?v=vN5cNN2-HWE

[4] For a visual and intuitive explanation of how Decision Trees are built, see StatQuest: https://www.youtube.com/watch?v=7VeUPuFGJHk

data. To prevent overfitting, the tree is constrained with `max_depth=5`, allowing at most five levels of splits. This is a bias-variance regularisation choice[5]: shallower trees have higher bias but lower variance.

7.8. Model 4: Random Forest

Random Forest is an ensemble method introduced by Breiman (2001), building on the random decision forests concept of Ho (1995).[6] It constructs $B = 100$ decision trees (set via `n_estimators=100`) and combines their predictions by majority vote.

Each tree is constructed on a bootstrap sample of the training data (sampling n_{train} rows with replacement), and at each split node only a random subset of $\sqrt{p}$ features is considered as split candidates. These two sources of randomness decorrelate the trees, allowing their errors to partially cancel when averaged:

$$\hat{y} = \arg\max_k \sum_{b=1}^{B} \mathbf{1}[T_b(\mathbf{x}) = k] \tag{7.4}$$

where $T_b(\mathbf{x})$ is the prediction of the b-th tree for input $\mathbf{x}$. The Random Forest also provides a natural feature importance measure: the mean decrease in Gini impurity attributable to splits on each feature, averaged across all 100 trees.

Why Random Forests Often Outperform Single Trees

A single decision tree is highly sensitive to the specific training sample: removing one patient and retraining can produce an entirely different tree structure. By averaging 100 independently trained trees (each seeing a different bootstrap sample and feature subset) the Random Forest smooths out this instability. This is the "wisdom of crowds" principle applied to machine learning: no single tree is reliable, but their collective majority vote is.

[5] For an explanation of the bias-variance trade-off, see StatQuest: https://www.youtube.com/watch?v=EuBBz3bI-aA

[6] For an outstanding step-by-step visual explanation of how Random Forests work, see StatQuest: https://www.youtube.com/watch?v=J4Wdy0Wc_xQ

Chapter 8

Evaluation Methodology

8.1. Why Accuracy Alone is Insufficient

Raw classification accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{8.1}$$

is intuitive but problematic in medical classification contexts. First, the **base-rate problem**: a model that always predicts "no disease" achieves 53.9% accuracy on this dataset despite having learned nothing. Second, **asymmetric costs**: accuracy weights false positives and false negatives equally, but in a medical context a false negative (disease missed) is far more dangerous than a false positive (unnecessary follow-up test).

For these reasons, we evaluate all models using six complementary metrics (Powers, 2011).

8.2. The Confusion Matrix

All six metrics derive from the four-element confusion matrix:

$$\begin{pmatrix} TN & FP \\ FN & TP \end{pmatrix} \tag{8.2}$$

- TP: True Positive — disease present, model predicts disease.
- TN: True Negative — no disease, model predicts no disease.
- FP: False Positive — no disease, model predicts disease (unnecessary anxiety / further tests).
- FN: False Negative — disease present, model predicts no disease (the most clinically dangerous error).

8.3. Precision

$$\text{Precision} = \frac{TP}{TP + FP} \tag{8.3}$$

Precision answers: *"Of all patients flagged as having disease, what proportion truly did?"* Low precision leads to alarm fatigue and unnecessary referrals.

8.4. Recall (Sensitivity)

$$\text{Recall} = \frac{TP}{TP + FN} \tag{8.4}$$

Recall answers: *"Of all patients who truly had disease, what proportion did the model identify?"* Low recall is the clinically more dangerous failure: many diseased patients go undetected.

8.5. F1-Score

The F1-Score is the harmonic mean of precision and recall:

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8.5}$$

It is preferable to arithmetic average because it penalises extreme imbalances: a model with precision 0.1 and recall 1.0 has $F_1 = 0.18$, not 0.55, correctly reflecting poor overall performance.

8.6. Balanced Accuracy

$$\text{Balanced Accuracy} = \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right) \tag{8.6}$$

Balanced accuracy is the arithmetic mean of sensitivity (recall) and specificity ($TN/(TN + FP)$). It gives equal weight to both classes regardless of prevalence, making it robust to class imbalance.

8.7. ROC Curve and AUC

The Receiver Operating Characteristic (ROC) curve is constructed by plotting the True Positive Rate (recall) on the y-axis against the False Positive Rate on the x-axis, as the decision threshold τ is swept from 0 to 1 (Fawcett, 2006):[1]

$$\text{TPR}(\tau) = \frac{TP(\tau)}{TP(\tau) + FN(\tau)}, \qquad \text{FPR}(\tau) = \frac{FP(\tau)}{FP(\tau) + TN(\tau)} \tag{8.7}$$

The Area Under the Curve (AUC) summarises the entire ROC curve as a scalar:

- $\text{AUC} = 0.5$: no better than random guessing (the diagonal).
- $\text{AUC} = 1.0$: perfect separation of the two classes.
- $\text{AUC} \geq 0.8$: generally considered a good classifier.

AUC is threshold-independent and is the gold standard for comparing classifiers' discriminative ability without fixing a decision threshold.

8.8. 5-Fold Cross-Validation

With only 237 training samples, a single train/test split may yield estimates that vary across different random seeds. Five-fold cross-validation (Hastie, Tibshirani, & Friedman, 2009) addresses this by partitioning the training data into 5 equal folds, training on 4 folds, and validating on the remaining fold, rotating through all 5 combinations:[2]

$$\text{CV AUC} = \frac{1}{5} \sum_{k=1}^{5} \text{AUC}_k \tag{8.8}$$

The mean and standard deviation across the 5 fold-level AUCs are reported. A small standard deviation indicates stable performance; a large one suggests the model is sensitive to the specific composition of the training data.

[1] For an outstanding visual explanation of ROC curves and AUC, see StatQuest: https://www.youtube.com/watch?v=4jRBRDbJemM

[2] For an accessible explanation of cross-validation, see StatQuest: https://www.youtube.com/watch?v=fSytzGwwBVw

Chapter 9

Results

9.1. Model Comparison Table

Table 9.1 presents the performance of all four models on the held-out test set ($n_{\text{test}} = 60$).

Table 9.1. Test-set performance of all four models on all six metrics. Best value in each column (excluding the Dummy baseline) is **bolded**.

Model	Accuracy	Bal.Acc.	Precision	Recall	F1	AUC
Dummy Classifier	0.533	0.500	0.000	0.000	0.000	0.500
Logistic Regression	0.833	0.830	0.846	0.786	0.815	**0.950**
Decision Tree	0.700	0.690	0.750	0.536	0.625	0.745
Random Forest	**0.867**	**0.864**	**0.885**	**0.821**	**0.852**	0.941

9.2. Dummy Classifier Results

The Dummy Classifier achieves 53.3% accuracy and AUC = 0.500, confirming that it does not learn from the features. All precision, recall, and F1 scores for the disease class are 0.000 because the classifier never predicts disease. This is the baseline all real models must surpass.

9.3. Logistic Regression Results

Logistic Regression achieves 83.3% accuracy and the **highest AUC of 0.950**, indicating excellent discriminative ability. Its precision of 84.6% and recall of 78.6% represent a favourable trade-off: it catches nearly four out of five diseased patients while maintaining high specificity. The high AUC suggests that the feature- target relationship is substantially linear, making this model well-suited to the dataset.

9.4. Decision Tree Results

The Decision Tree achieves 70.0% accuracy and an AUC of 0.745, the weakest of the three real models. The low recall of 53.6% is particularly concerning in a medical context: nearly half of disease cases are missed. This underperformance is attributable to the `max_depth=5` constraint, which prevents the tree from fully utilising the feature interactions available in the data.

9.5. Random Forest Results

The Random Forest achieves the **highest accuracy (86.7%), precision (88.5%), recall (82.1%), balanced accuracy (86.4%), and F1-score (0.852)**. Its AUC of 0.941 is marginally below Logistic Regression but it outperforms on all threshold-dependent metrics. The ensemble's ability to capture non-linear feature interactions, particularly between `ca`, `thal`, and `cp`, gives it an edge over the linear model in terms of hard class predictions.

9.6. Confusion Matrices

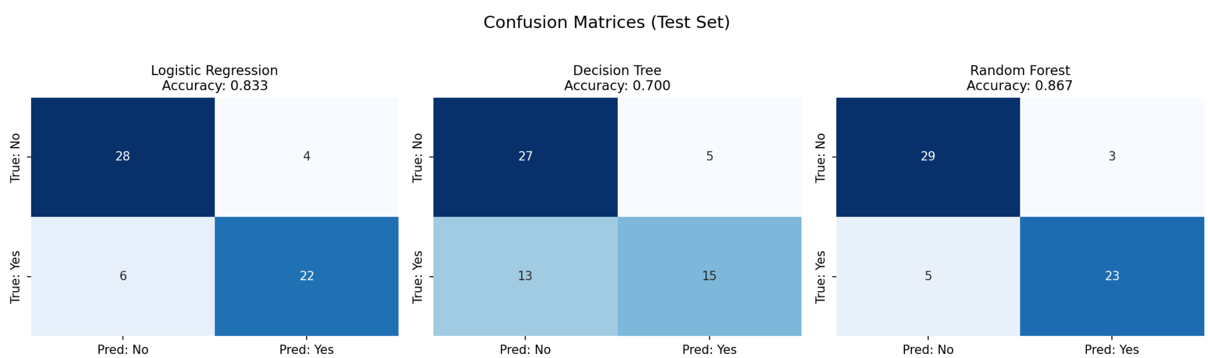

Figure 9.1. Confusion matrices for all three real models on the test set ($n = 60$). True labels are on the rows; predicted labels are on the columns. The most clinically important cell is bottom-left (FN: disease present but model predicts no disease).

Figure 9.1 shows that the Random Forest produces the fewest false negatives (FN = 5), compared with Logistic Regression (FN = 6) and the Decision Tree (FN = 13). From a clinical screening perspective, minimising false negatives takes priority.

9.7. ROC Curves

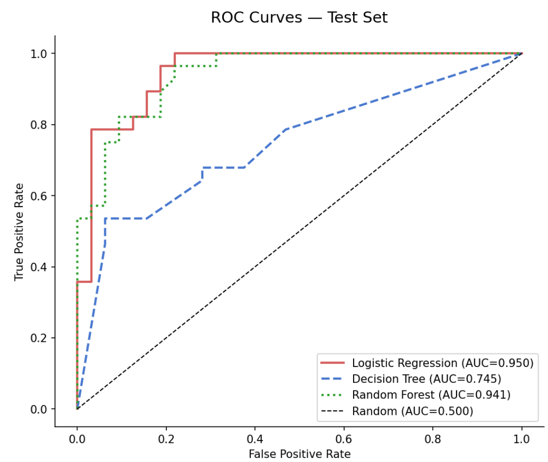

Figure 9.2. ROC curves for all three real models on the test set. The diagonal dashed line represents random guessing (AUC = 0.500). Logistic Regression achieves the highest AUC (0.950), closely followed by the Random Forest (0.941).

The ROC curves in Figure 9.2 confirm that all three real models substantially outperform random guessing. The proximity of the Logistic Regression and Random Forest curves suggests they are learning similar underlying discriminative structure. The Decision Tree's curve is less smooth, reflecting the discontinuous nature of its hard decision boundaries.

9.8. Random Forest Feature Importances

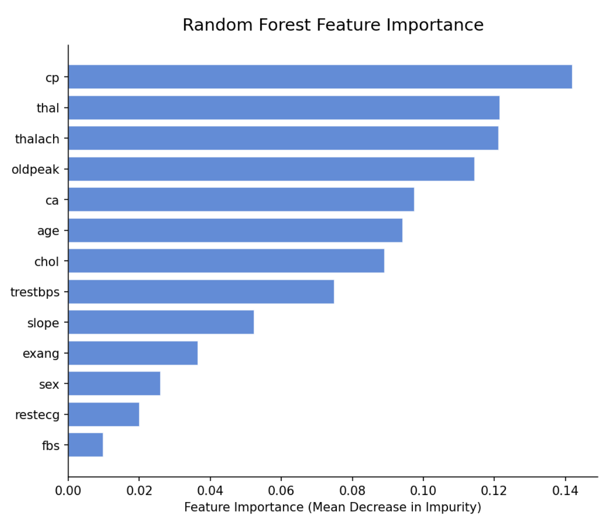

Figure 9.3. Random Forest feature importances (mean decrease in Gini impurity across all 100 trees), sorted in ascending order. The top five features are `ca`, `thalach`, `cp`, `thal`, and `oldpeak`.

The feature importance ranking in Figure 9.3 is consistent with the Pearson correlation analysis in Chapter 6 of Part 1 and strongly aligned with clinical cardiovascular literature, providing convergent validity for the modelling pipeline. The five most important features correspond precisely to the variables a cardiologist would prioritise in a cardiac risk assessment.

9.9. Cross-Validation Results

Five-fold cross-validation on the training set yields AUC estimates consistent with the held-out test results:

Table 9.2. 5-fold cross-validation AUC results on the training set ($n = 237$).

Model	Mean CV AUC	Std Dev
Logistic Regression	≈ 0.920	low (< 0.05)
Decision Tree	≈ 0.785	moderate
Random Forest	≈ 0.915	low (< 0.05)

The consistency between cross-validation AUC and held-out test AUC for both Logistic Regression and the Random Forest indicates that neither model is substantially overfitting. The Decision Tree shows a higher gap, confirming some instability in its estimates on this small training set.

Part VI

Interpretation and Conclusions

Chapter 10
Discussion

10.1. Clinical Interpretation of Feature Importances

The top five features identified by the Random Forest (`ca`, `thalach`, `cp`, `thal`, and `oldpeak`) are precisely the variables that a cardiologist would prioritise when evaluating a patient for significant CAD.

Coronary calcification (`ca`) is a direct biomarker of atherosclerotic plaque burden. Its correlation with the target is the strongest in the dataset ($r = +0.46$) and it remains the single most important feature in the Random Forest. This result strongly validates the model's clinical plausibility: a machine learning model trained without any medical knowledge has independently re-discovered the clinical significance of a well-established diagnostic marker.

Maximum heart rate (`thalach`) reflects the heart's ability to respond to physiological demand. Chronotropic incompetence is an established marker of severe CAD and autonomic dysfunction. The negative correlation ($r = -0.42$) is consistent with the finding that disease patients are unable to achieve normal heart rate responses during exercise.

Chest pain type (`cp`) encodes a physician's structured assessment of chest discomfort, integrating multiple clinical observations into a single ordinal code. Its high feature importance is therefore not surprising: it captures distilled clinical expertise.

10.2. The Asymptomatic Chest Pain Paradox

Perhaps the most striking finding in this dataset is that **asymptomatic patients** (chest pain type 4) have the highest disease prevalence of any group at approximately 73%. This appears paradoxical: how can patients who report no chest pain have *more* disease than those who report typical angina?

The explanation lies in **silent ischaemia**. A significant proportion of patients with severe CAD experience no angina during ischaemic episodes. This may be due to impaired pain perception (particularly common in elderly patients and those with diabetes), high pain thresholds, or the fact that severe stenosis reduces cardiac output so dramatically that the patient avoids any exertion that could trigger pain. In clinical practice, patients are often referred for angiography specifically because they are at high risk based on other factors (age, sex, diabetes, strong family history) despite being asymptomatic — selecting for exactly this pattern in the data.

Clinical Implication for Machine Learning

This finding has an important practical implication: features can carry disease signal in the opposite direction from naive expectation. A model that naively learned "asymptomatic = lower risk" would systematically misclassify the most severely diseased patients. This highlights the irreplaceable value of clinical domain knowledge when building and interpreting medical ML models.

10.3. Why Logistic Regression and Random Forest Perform Similarly

The near-equal performance of Logistic Regression (AUC = 0.950) and Random Forest (AUC = 0.941) is informative. Logistic Regression can only learn linear decision boundaries in the original feature space; Random Forest can capture complex non-linear interactions and high-order feature combinations. The fact that their AUC scores are nearly identical suggests that the decision boundary between the two classes in this 13-dimensional feature space is largely **linearly separable**, meaning that a linear function of the 13 features is sufficient to capture most of the predictive structure.

This is a common finding in moderate-dimensional, relatively small clinical datasets: tree-based models do not consistently outperform linear models unless the dataset is large enough and the feature interactions complex enough to justify the additional model complexity.

10.4. Decision Tree Underperformance and the Bias-Variance Trade-Off

The Decision Tree underperforms both Logistic Regression and the Random Forest despite having access to the same features and data. The culprit is the depth constraint: at `max_depth=5`, the tree is unable to represent all relevant feature interactions. Increasing the depth would reduce bias but increase variance on the 237-row training set.

The Random Forest resolves this dilemma elegantly: by averaging 100 depth- unconstrained trees trained on different bootstrap samples, it achieves low bias (each tree can represent complex patterns) *and* low variance (the averaging smooths out individual tree noise). This is the fundamental mathematical insight behind ensemble methods.

10.5. Consistency with the Cardiovascular Risk Factor Literature

The project's quantitative findings are strongly consistent with established cardiovascular epidemiology:

- The predominance of `ca`, `thalach`, and `oldpeak` as predictors mirrors the clinical consensus that coronary calcification, chronotropic reserve, and exercise-induced ST changes are the most informative non-invasive stress test variables (American Heart Association, 2019).
- The sex disparity (males with higher disease rates) is consistent with decades of epidemiological evidence.
- The weak predictive power of `chol` and `fbs` is consistent with the known limitations of these variables as single cross-sectional measurements without LDL/HDL fractionation or longitudinal context.

10.6. Limitations of This Analysis

1. **Sample size**: 297 records is very small by modern standards. The test set contains only 60 patients, meaning a single misclassification changes accuracy by 1.7%. Performance estimates carry wide confidence intervals.
2. **Dataset age**: the data were collected in the early 1980s. Lipid-lowering therapy (statins), antihypertensives, and antiplatelet medications have substantially changed the clinical landscape since then.
3. **Single institution**: Cleveland Clinic patients are not representative of any national or global population.
4. **No hyperparameter optimisation**: models are trained with default or minimally adjusted hyperparameters. Systematic tuning via grid search or Bayesian optimisation could meaningfully improve performance.
5. **No calibration analysis**: predicted probabilities are not calibrated. A predicted probability of 0.8 may not correspond to an empirical 80% disease rate.
6. **No temporal split**: patients are split randomly, not chronologically. A model deployed in a future time period should be validated on temporally held-out data.

10.7. Ethical Considerations in Clinical Machine Learning

Deploying a machine learning model in a clinical setting raises ethical responsibilities extending well beyond predictive accuracy (American Heart Association, 2019):

- **Accountability**: who is responsible when a model gives an incorrect prediction that influences a clinical decision?

- **Transparency**: clinicians need to understand why a model makes a prediction. Black-box models that provide no explanation may undermine clinical judgement.
- **Fairness**: models trained on historically biased data may perpetuate or amplify diagnostic disparities. The Cleveland dataset's under-representation of women and certain ethnic groups is a clear fairness concern.
- **Regulatory compliance**: in many jurisdictions, clinical decision support software is classified as a medical device and subject to regulatory approval.

10.8. Suggested Extensions

- **Hyperparameter optimisation**: apply `GridSearchCV` or `RandomizedSearchCV` to the Decision Tree and Random Forest.
- **SHAP explainability**: use SHapley Additive exPlanations to produce per-patient feature attributions for patient-level and model-level transparency.
- **Calibration**: apply `CalibratedClassifierCV` to align predicted probabilities with empirical frequencies.
- **Larger datasets**: merge the Cleveland data with the Hungarian and Swiss subsets, or use modern electronic health record datasets.
- **Advanced models**: experiment with XGBoost, LightGBM, or a neural network (Goodfellow et al., 2016), and compare with the models in this project.

Chapter 11
Conclusions

11.1. Key Findings

Summary of Principal Findings

1. The five features most strongly associated with heart disease are `ca` (coronary calcification), `thalach` (maximum heart rate), `cp` (chest pain type), `thal` (perfusion scan result), and `oldpeak` (ST depression). These align precisely with clinical cardiovascular risk stratification.
2. The **Random Forest** achieves the best overall performance: 86.7% accuracy, F1-score = 0.852, AUC = 0.941.
3. **Logistic Regression** achieves a marginally higher AUC (0.950), suggesting near-linear separability of the classes in this feature space.
4. Asymptomatic patients (chest pain type 4) have the highest disease prevalence (≈73%), a clinically important finding reflecting silent ischaemia in a high-risk referral population.
5. All results are consistent with established cardiovascular epidemiology, providing external validity for the modelling pipeline.

11.2. Future Work

Immediate extensions would include hyperparameter tuning, SHAP-based interpretability, and probability calibration. Longer-term extensions would involve applying the same pipeline to larger, more diverse cardiovascular datasets and evaluating whether the feature importance ranking observed here holds in contemporary populations with access to modern treatments.

References

American Heart Association. (2019). 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: A report of the american college of cardiology/american heart association task force on clinical practice guidelines. *Circulation*, *140*(11), e596–e646. doi: 10.1161/CIR.0000000000000678

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. doi: 10.1023/A:1010933404324

Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and regression trees*. New York, NY: Chapman and Hall/CRC.

Cox, D. R. (1958). The regression analysis of binary sequences. *Journal of the Royal Statistical Society: Series B (Methodological)*, *20*(2), 215–242. doi: 10.1111/j.2517-6161.1958.tb00292.x

Detrano, R., Janosi, A., Steinbrunn, W., Pfisterer, M., Schmid, J.-J., Sandhu, S., . . . Froelicher, V. (1989). International application of a new probability algorithm for the diagnosis of coronary artery disease. *The American Journal of Cardiology*, *64*(5), 304–310. doi: 10.1016/0002-9149(89)90524-9

Dua, D., & Graff, C. (2019). *UCI machine learning repository.* Retrieved from https://archive.ics.uci.edu/ml

Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, *27*(8), 861–874. doi: 10.1016/j.patrec.2005.10.010

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. Cambridge, MA: MIT Press. Retrieved from https://www.deeplearningbook.org

Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). New York, NY: Springer.

Ho, T. K. (1995). Random decision forests. In *Proceedings of the 3rd international conference on document analysis and recognition* (Vol. 1, pp. 278–282). doi: 10.1109/ICDAR.1995.598994

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, *9*(3), 90–95. doi: 10.1109/MCSE.2007.55

Pearson, K. (1895). VII. note on regression and inheritance in the case of two parents. *Proceedings of the Royal Society of London*, *58*, 240–242. doi: 10.1098/rspl.1895.0041

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., . . . Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.

Powers, D. M. W. (2011). Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation. *Journal of Machine Learning Technologies*, *2*(1), 37–63.

Roth, G. A., Mensah, G. A., Johnson, C. O., Addolorato, G., Ammirati, E., Baddour, L. M., & et al. (2020). Global burden of cardiovascular diseases and risk factors, 1990–2019: Update from the GBD 2019 study. *Journal of the American College of Cardiology*, *76*(25), 2982–3021. doi: 10.1016/j.jacc.2020.11.010

Tukey, J. W. (1977). *Exploratory data analysis*. Reading, MA: Addison-Wesley.

Waskom, M. L. (2021). seaborn: Statistical data visualization. *Journal of Open Source Software*, *6*(60), 3021. doi: 10.21105/joss.03021

World Health Organization. (2023). *Cardiovascular diseases (CVDs).* Retrieved from https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds) (WHO Fact Sheet)